Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/01/2008
PI Name:	Lupton, Joanne R. Ph.D.		
Project Title:	Nutritional Countermeasures to Radiation-Enhanced Colon Cancer		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas Engineering Experiment Station) Chapkin, Robert (Texas A&M University) Ford, John (Texas A&M University) Turner, Nancy (Texas A&M University) Carroll, Raymond (Texas A&M University)		
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Task Description:	1. Original Aims. The primary goal of this project is to develop a set of colon cell gene expression profiles that accurately characterize the radiation-enhancing effect on colon tumor development and the chemoprotective effect of dietary countermeasures in this process. This goal is addressed by specific aim #1: From the global transcriptional profile (analyzed by microarray technology) in samples isolated from colonic mucosa, determine which sets of gene expression profiles best characterize the initiation stage, the promotion stage (aberrant crypt development) and the final tumor stage of colon cancer development using a 2 x 4 x 4 experimental design (+ or – radiation; four time points and 4 diets). The secondary goal, addressed by specific aim #2, is to compare gene expression profiles developed from exfoliated cells to those from colonic mucosal cells to determine how well they reflect in vivo events. This information is important for our short-term future goal of using fecal material from humans as a noninvasive method to monitor changes in gene expression patterns over time. Our long-term future goal is to have sets of gene expression profiles that can be used to monitor astronauts before, during, and after space flight to detect exposure and response to a radiation event so that appropriate intervention strategies (e.g., dietary countermeasures) can be implemented as needed. Data generated from this research program will support the NASA/NSBRI goals of: 1) gaining greater understanding of the mechanisms involved in the synergy of radiation and other environmental insults on carcinogenesis, 2) developing a system to monitor the need for countermeasure administration during or after flight, and 3) development of countermeasures that can be applied before, during and after flight. 2. Key Findings. Analysis of fecal samples collected over time indicated there were a variety of genes with differences in expression in response to diet, radiation treatment, stage of disease progression and their interactions. Radiation exposure caused changes in expression for 125 genes; time of sampling affected the expression of 369 genes; and diet caused differential expression in 387 genes. Of the differentially expressed genes with a known function in the two radiation exposure groups, there were diverse functional activities represented. Expression of genes involved in metabolism, regulation of cell proliferation and differentiation, signaling cascades, etc. in colonocytes were affected by radiation exposure. Some of the genes are involved in basic metabolic pathways, such as, cholesterol biosynthesis while others are members of key regulatory signaling cascades or are regulators of gene transcription, that are known to be involved in carcinogenesis (CREBBP, histone deacetylase, MAP kinase phosphatase). Diet influenced a variety of signaling pathways through changes in genes such as parathyroid hormone receptor and G-protein coupled receptor 54. Genes, such as Notch 3, were affected by the stage of tumorigenesis. After pathway analyses, many genes were identified that were over-represented for their respective biological processes (genes involved in the metabolism of carbohydrates and lipids, ion transport and signaling pathways). There were more than 300 genes for which differences in gene expression were observed for the diet*time interaction, indicating the transformations occurring in the colon during tumor development were sensitive to dietary intervention. Examples of the functional categories involved are ATP binding, glutathione synthesis, metabolic rate regulation, cytokine activity and oxidoreductase activity. These proteins are important in everything from buffering cellular redox status, immune function, metabolic rate regulation, metabolism of chemicals and signaling through protein kinase pathways. The greatest amplitude of change in expression identified by clustering analysis was noted for the 28-week samples from the irradiated rats consuming corn oil/cellulose diets, which is the diet most promotive of colon cancer development. These results have been presented at the NASA Human Research Program Investigator's Workshop. February 11-14, 2007, Houston, TX; Humans in Space 07 Conference, May 22, 2007, Beijing, China; and 18th Annual NASA Space Radiation Investigators' Workshop, July 13-15, 2007, Rohnert Park, CA. 3. Impact of Findings on Project Goals. The observations made to date support the feasibility of using fecal polyA+ RNA as a non-invasive measure of colonocyte gene expression in Earth-based medicine and to monitor in-flight radiation exposure and cell responses to the insult. It would therefore be possible to identify interventions for space flight personnel in order to slow disease progression until their return to Earth. This work also indicates dietary modifications influence gene expression in colonocytes and the pattern of changes in expression that occur during disease progression. It is therefore possible to use diet as a mediator in colon carcinogenesis, so as to reduce the enhancement of cancer risk posed by exposure to galactic cosmic radiation. Importantly, the observations made over time have detected strong interactions in treatments over time suggesting that the full effect of a treatment will not likely be understood by a single sample collected at a specific reference time point. 4. Proposed Research Plan for the Coming Year. We will complete the fecal sample analyses in the next 3 months and initiate the statistical analyses for those samples. Results from the fecal and mucosal microarray analyses will be statistically analyzed and key genes identified so that the results may be validated through PCR assays. We are preparing publications from the first set of analyses and will generate manuscripts from the second round of fecal analyses and the corresponding mucosal sample this year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary deterrent to great reductions in the number of deaths from colon cancer is getting individuals to undergo screening. Data generated by this project will extend to Earth-based medical care by providing a system of detection that is non-invasive and should facilitate patient compliance with screening recommendations. Development of a set of gene expression profiles that are indicative of colon cancer development, and that can be monitored through a noninvasive procedure would greatly increase screening success and reduce the current high rates of morbidity and mortality associated with this disease.
Task Progress:	This project aims to understand the changes in gene expression that occur in the colon after exposure to radiation, followed by injection of a chemical carcinogen (the chemical carcinogen is a model of sporadic colon cancer in humans). One goal is to understand disease progression and to identify gene expression patterns that are indicative of the disease stage. Our second goal is to compare gene expression profiles of mucosal samples to samples of RNA from sloughed cells in order to develop a non-invasive procedure of detecting the changes. Lastly, we wanted to understand how diets containing either corn oil or fish oil as the lipid component and either cellulose or pectin as the fiber source influence the response to radiation and carcinogen exposure, and thus disease progression. Over the last year we have statistically processed the output of the microarray analyses from the mucosal samples (n = 212). Secondly, we are in the final stages of collecting the microarray data from the fecal samples collected at the same time as the mucosal samples. Analyses of fecal samples collected at three times after carcinogen injection (7, 14 or 28 weeks) identified genes affected by diet, radiation exposure, and stage of disease, as well as their interactions. Radiation exposure caused changes in the expression of 125 genes; 48 genes were up regulated and 77 genes were down regulated by radiation

	exposure. Some of these genes are involved in transcriptional regulation and chromatin remodeling, making them critical determinants of disease development through their influence on the expression of other genes and cell proliferation. Diet influenced the response to radiation by influencing the expression of 95 genes, of which some are involved in metabolism, as well as intestinal growth and nutrient sensing. The greatest shift in patterns of gene expression were observed in rats consuming corn oil and cellulose and exposed to radiation, especially at the 28-week sample, which is near the tumor end point. Gene ontology analysis of these significant genes yielded several genes involved in carbohydrate and lipid metabolism and biosynthesis. We are in the process of identifying the patterns of gene expression in these time course samples that are most predictive of the tumor results.
Bibliography Type:	Description: (Last Updated: 08/22/2020)
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